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Integrated taxiing and TLOF pad scheduling using different surface directions with fairness analysis

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Abstract—
Vertical Take-Off and Landing (VTOL) vehicles have gained immense popularity in the delivery drone market and are now being developed for passenger transportation in urban areas to efficiently enable Urban Air Mobility (UAM). UAM aims to utilize the urban airspace to address the problem of heavy road congestion in dense urban cities. VTOL vehicles require vertiport terminals for landing, take-off, refuelling (or charging), and maintenance. An efficient scheduling algorithm is essential to maximize the throughput of the vertiport terminal while maintaining safety protocols to handle the UAM traffic. While traditional departure and taxiing operations can be applied in the context of vertiport terminals, specific algorithms are required for take-off and landing schedules. Unlike fixed-wing aircraft that require a runway to take-off and climb in a single direction, VTOL vehicles can approach and climb in several directions. We propose a Mixed Integer Linear Program (MILP) formulation to schedule flights for taxiing and climbing in multiple directions after take-off. Our results show that increasing the number of climbing directions can reduce delays by over 50%. We also introduce a fairness metric to ensure a uniform delay distribution.

I. Introduction

Future modes of transportation in domestic mobility are expected to integrate Urban Air Mobility (UAM) with surface transport as frequent traffic congestion and rerouting can cause unpredictable delays. UAM is expected to provide cost-effective air travel with fully automated and highly efficient operational procedures [1]. UAM vehicles, primarily rotary wing structure, are equipped with the ability to take-off and land in vertical spaces called "Vertiports". Traffic management of UAM vehicles, along with other Unmanned Aerial Systems (UAS) such as delivery drones, are governed by UAS Traffic management or UTM. UTM mandates that UAM vehicles should land and take-off only at specified vertiports constructed within cities. Several international aviation bodies such as EASA [2] and FAA [3] have recently published design guidelines for the construction of vertiports.

Some analogy can be drawn between UTM and conventional Air Traffic Management (ATM). The UTM traffic handling capability and demand in a city can be compared to busy airports in metro cities. The vertiport terminal will be the bottleneck for UTM under high demand conditions that might result in congestion and wastage of associated resources such as fuel (energy), manpower, and time;leading to uneconomical operation. Hence, optimizing vertiport operations is vital in intelligent transportation research, and several approaches have already been investigated for airport ground operations ([4], [5]) that can be applied to vertiport terminals. Unlike traditional aircraft where take-off and climb is a bottleneck, VTOL aircraft use TLOF (Touchdown and Lift-Off) pads to take-off and can climb in any direction. This simple operation, of climbing in any direction, can be exploited to design an appropriate schedule to increase a vertiport terminal's throughput. The details about VTOL vehicle operations are described in Section III and depicted in Figure 3.

In this paper, we formulate a scheduling problem using the concept of multiple climb directions to optimise vertiport terminal operations. The scheduling problem formulation leads to a Mixed Integer Linear Program (MILP). The results obtained from the solution of the MILP formulation are analysed to show the impact of the number of climb directions and the number of flights on the delay incurred. We introduce a fairness metric to ensure a uniform delay distribution for UAM vehicles.

II. Related Work

Taxiing and runaway scheduling is a well-researched area for traditional aircraft and airports. Numerous works like ([6],[7] and [8]) have developed genetic algorithms for solving taxiing problems which calculates a fit function. The thesis work of Simaiakis [9] analyses the departure process of an airport using queuing models and proposes dynamic programming algorithms. Some latest work like [10], [11] and [12] also consider the gate assignment problem as an essential part of taxiing route and airport scheduling problem.

Recent works on vertiport scheduling and design like [13] present the methods to calculate optimal "Required time of arrival" by MILP considering energy and flight dynamics and [14] proposes a multi-ring structure over multi-vertiport.The multi-ring structure exists outside the vertiport airspace and is not handled by the vertiport. The thesis work presented in [15] talks about the tools that can be used for vertiport design simulation and analysis. Since a vertiport can have multiple gates and TLOF pads, multiple works like [16], [17], and[18] have done extensive studies on the design of vertiport, its capacity and throughput analysis.

To the best of our knowledge, no work has considered multiple climb directions for take-off, along with taxiing on the surface of a vertiport terminal. The optimisation problem proposed in [19], which we use as our base, provides queues at the runaway . As take-offs are free-flowing in the presence of multiple climbing directions and with the correct schedule

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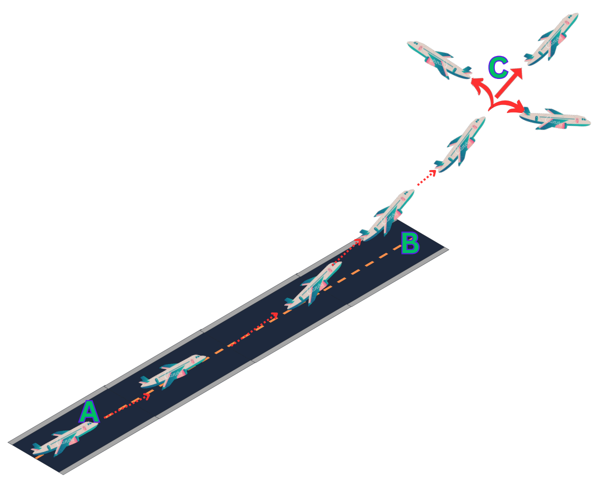
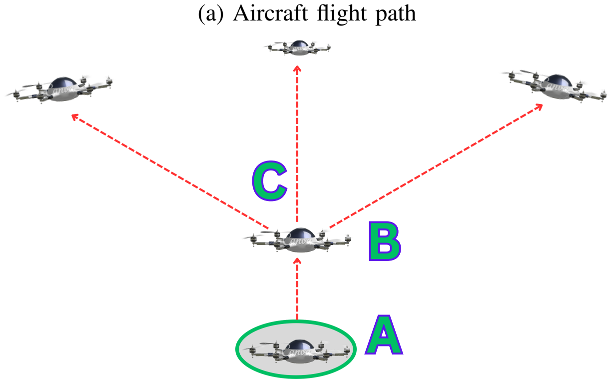
of aircraft, queues are not needed at the entrance of the TLOF pad. The impact of queuing, if any, is left for future study.

The paper presents the following contributions: (1) We have proposed a novel method where a UAM flight can climb using multiple directions. (2) We propose a MILP problem for taxiing and TLOF pad scheduling along with the direction of ascent. (3) We have studied the delay performance of flights with increasing number of climb directions. (4) We proposed a fairness metric to ensure uniform delay distribution for flights and compare the results of two methods of applying the metric.

III. Vertiminal Scheduling Problem



(a) Aircraft flight path



(b) Drone flight path

Fig. 1: Comparison of flight paths

Our goal is to minimise departure delays such that throughput can be maximised. Figure 1a shows a conventional airport system where an aircraft will enter the runway, accelerate to the required speed, take-off and deviate in the direction of the approved flight plan after attaining a certain altitude. The two line segments "A to B" and "B to C" depict the common distance occupied by only one flight at a time, thus being a bottleneck to the departing traffic.

We define **vertiminal** as a vertiport terminal, which may be regarded as similar to a bus terminal. A vertiminal handles traffic of **UAM VTOL capable aircraft**. Such aircraft are defined as heavier-than-air aircraft, other than an aeroplane or helicopter, capable of performing VTOL using more than two lift or thrust units that are used to provide lift during the take-off and landing [2]. A UAM aircraft departs from the gate and travels a distance, called a **taxiway**, before reaching the TLOF pad. Figure 1b shows a VTOL vehicle take-off from a **TLOF pad**. The line segment "A to B" shows the vertical take-off and unlike in Figure 1a, the line segments "B to C" is of 0 length. Unlike conventional runways, these pads are compact, and thus a VTOL vehicle can reach cruising altitude (point C) quickly and start its approved flight plan. Back-to-back take-offs from TLOF pads use the same path. Figure 2 shows the **Obstacle Free Volume** (OFV), which is a funnel-shaped area with 2 climb direction surfaces present on the TLOF pad. It guarantees that VTOLs can accomplish take-offs and landings within a sizable vertical segment, allowing them to account for environmental and noise limits in urban settings. At the end of OFV boundary, there can be multiple climb direction surfaces for VTOL vehicles to climb to the **vertiexit**; an altitude where UAM VTOL aircraft exits from the vertiminal airspace. While there is a minimum separation time required for successive take-offs due to wake turbulence on the TLOF pad, we have the opportunity to reduce the length of the shared path (bottleneck). Two aircraft moving in the same surface direction must be separated by the minimum separation distance. However, if the two aircraft move in different surface directions, no minimum separation distance needs to be maintained. Hence, with more than one surface direction, the total delay can be reduced compared to that with a single surface direction. The addition of multiple directions thus effectively reduces shared path length.

A departure-ready flight can experience delays in its

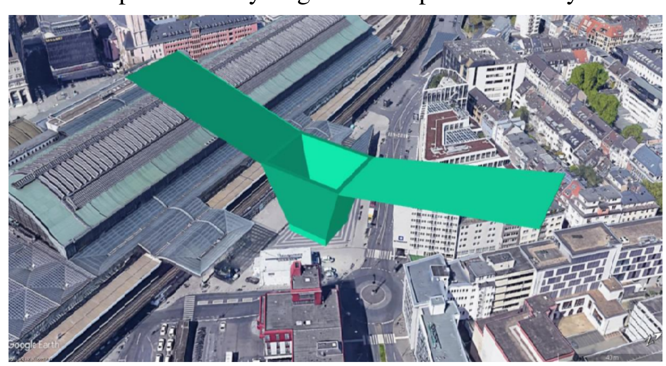


Fig. 2: Example of OFV with 2 climb direction surfaces [2].

path due to the following: (a) Separation requirements on the taxiways; (b) TLOF pad clearance while the previous VTOL aircraft climbs the OFV boundary; (c) Wake turbulence requirements on TLOF pad; (d) Separation requirements on the climbing surface in the case of multiple UAM vehicles taking the same direction.

A. Problem Formulation

1) Assumptions: We will use the term "surface direction" throughout to denote the take-off climb direction surface. We will use the term "VTOL" instead of "UAM VTOL capable aircraft" throughout. As explained below, several assumptions are made for simplifying the modelling of

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Vertiminal Parameters	
N_R	the set of nodes representing all TLOF pads at the vertiminal.
$N_O^{n_r}$	The set of nodes representing OFV boundary on a TLOF pad $n_r \in N_R$
$N_F^{n_r}$	The set of nodes at the vertiexit from TLOF pad $n_r \in N_R$
$N_O = \bigcup_{n_r \in N_R} N_O^{n_r}; \quad N_F = \bigcup_{n_r \in N_R} N_F^{n_r}$	
N_T	the set of all nodes n on the ground where a node could be the intersection of two taxiways, an entrance or exit to a taxiway or a TLOF pad.
N	$N_T \cup N_R \cup N_O \cup N_F$
L_T	set of links connecting nodes $n_t \in N_T$
L_V	set of virtual links or zero-length links connecting nodes $n_t \in N_T$ which are entrance to TLOF pads
L_O	set of links connecting nodes $n_r \in N_R$ *and* $n_o \in N_O^{n_r}$
L_F	set of links connecting nodes $n_o \in N_O^{n_r}$ *and* $n_f \in N_F^{n_r}$ $\forall n_r \in N_R$
L	$L_T \cup L_V \cup L_O \cup L_F$
$l_{a,b}$	$\in L$ is a link connecting nodes a and b also represented as (a,b)
G	(N, L), the vertiminal network graph
VTOL sets and Parameters	
A	the set of all departing VTOLs.
A^o	the set of all departing VTOLs plus one dummy VTOLs 0.
$\mathtt{DRGATE_i}$	$\forall i \in A$ the time at which a departing flight i is ready to leave the gate after passenger boarding.
A^n	$\forall n \in N$: the set of all VTOLs whose route passes through node n.
A_0^n	$\forall n \in N$: the union of the set of all VTOLs whose routes pass through node n and the set of the dummy VTOLs, i.e., $A_0^n : A^n \cup \{0\}$
$\tau(i)$	$\forall i \in A$: mapping function which gives the TLOF pad $n_r \in N_R$ assigned to a VTOL i.
$\Lambda(i)$	$\forall i \in A$: mapping function which gives surface direction $l^i_{n_f, n_o} \in L_F, n_f \in N_F^{n_r}, n_o \in N_O^{n_r}$ assigned to a VTOL i from TLOF pad $n_r \in N_R$
$\Lambda^F(i)$	$\forall i \in A$: mapping function which gives vertiexit node $n_f \in N_F$ on surface direction $\Lambda(i)$
$\Lambda^O(i)$	$\forall i \in A$: mapping function which gives OFV boundary node $n_o \in N_O$ on surface direction $\Lambda(i)$
W_{ij}^{tsep}	$\forall i, j \in A, i \neq j$: required safe separation time at landing or takeoff (calculated from a distance, if necessary) of a VTOL i from its immediate trailing a VTOL j

TABLE I: Sets and notation used in objective problem

vertiminal operations.
(1) Vertiminals will provide a standard surface direction of fixed width to ensure VTOL remain within this specified limit. (2) The climbing speed of a VTOL is assumed to be given and governed by equipment type, environment and regulations. (3) Vertiminals have standard taxi routes. Therefore, given a TLOF pad and a gate, the taxi route for a VTOL is predefined. (4) Nominal taxi speed in is assumed to be given. Therefore, given the length of the taxiway, the minimum and maximum travel time can be estimated. (5) The passage times at critical points along taxi routes, OFV and surface direction given by the solution to the optimisation problem can be met by a VTOL. (6) Gates can be modelled as a source for flights, and vertiexit behaves as a sink for flights.

2) Input Parameters: Table I shows the sets (VTOLs, nodes, links etc.), notation and configuration parameters used in the objective formulation.

VTOL Operations: Figure 3 represents a general path for a departing VTOL from gate to vertiexit. Let P_i $\forall i \in A$ represent the physical taxi route for a VTOL i, from its first node to its last node. P_i can be represented by an ordered set of k_i nodes $\{n_1^i, n_2^i, ..., n_{k_i}^i\} \in N_T$ starting with the first node n_1^i of the route (gate exit for $i \in A$), progressing through the intermediate nodes and ending at the last node of route $n_{k_i}^i$ which is the assigned TLOF pad entrance for $i \in A$, where k_i denotes the total number of nodes involved in P_i. The given route for a departure VTOL consists of three parts: (a) P_i sequentially ordered set of nodes $n_t \in N_T$, from the starting gate node on the network G to the entrance of the holding space of the assigned TLOF pad; (b) Link $l_{n_o, n_r} \in L_O$ connecting TLOF pad $n_r \in N_R$ and OFV boundary $n_o \in N_O^{n_r}$; (c) Link $l_{n_o, n_f} \in L_F, n_f \in N_F^{n_r}, n_o \in N_O^{n_r}$ represents a climb surface direction to the assigned TLOF pad $n_r \in N_R$.

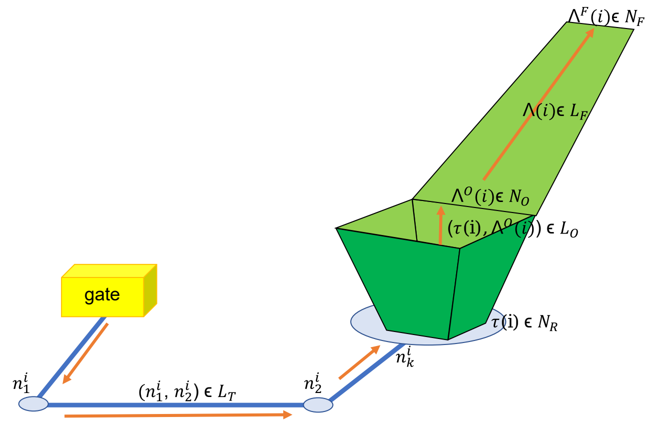

Fig. 3: The nodes and link representation for the optimisation problem explained in Section III-A.4. The orange line represents the path of a VTOL from gate to vertiexit.

W_g	Weight assigned to a departing VTOL for the time it expends at the gate. $0 \leq W_g \leq 1$.
W_t	Weight assigned to a departing VTOL for the taxiing time it expends. The requirement $W_g \leq W_t \leq 1$ ensures that a VTOL should spend more time at the gate rather than on taxiways to reduce congestion.
W_c	Weight assigned to a departing VTOL to the time it expends to reach vertiexit from the TLOF pad. $0 \leq W_c \leq 1$.

TABLE II: Weights used in objective function

3) Decision Variables: The decision variables used in the MILP formulation are shown in Table III.

t_i^n	$\forall i \in A, \forall n \in N$: (non-negative variable) the time at which VTOL i reaches node n. Note that the VTOL does not slow down, let alone stop before reaching a node and continues to achieve smooth travel unless the node is at the TLOF pad entrance.
x_{ij}^n	$\forall n \in N, \forall i, j \in A_0^n, i \neq j$ (binary variable, about immediate predecessor), $x_{ij}^n = 1$ if and only if VTOL i reaches node n immediately before VTOL j does, $x_{ij}^n = 0$ otherwise. If VTOL j is the first VTOL to reach node n, then set $x_{0j}^n = 1$, else 0. If VTOL i is the last VTOL to reach n, then set $x_{i0}^n = 1$, else 0.
y_{ij}^n	$\forall n \in N, \forall i, j \in A_0^n, i \neq j$ (binary variable, about predecessor), $y_{ij}^n = 1$ if and only if VTOL i reaches node n before VTOL j does; $y_{ij}^n = 0$ otherwise.

TABLE III: Decision variables

4) Objective Function: The total time spent by a departing VTOL from leaving the gate to reaching the vertiexit consists of three segments: (a) waiting time at the gate, (b) gate to the entrance of TLOF pad, (c) TLOF pad to vertiexit. All these delays are added in the objective function with the associated weights (TABLE II). The weights can be chosen depending on the relative importance of the three

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delay terms.

$$\sum_{i\in A}[W_g(t_i^{n_1^i} - \mathtt{DRGATE_i}) + W_t(t_i^{n_{k_i}^i} - t_i^{n_1^i}) + W_c(t_i^{\Lambda^F(i)} - t_i^{\tau(i)})] \tag{1}$$

C1: The constraint equation (2) is the minimum time by which the VTOL should leave the gate and start taxiing.

$$t_i^{n_1^i} \geq \mathtt{DRGATE_i} \quad \forall i \in A \tag{2}$$

C2: For speed control over taxiing ways and in surface direction and ensuring smooth travel, constraint equation (4) is required. The terms $T^{min}_{il^i_{p,p+1}}$ and $T^{max}_{il^i_{p,p+1}}$ are the respective minimum and maximum time taken by VTOL i on the link $l^i_{p,p+1} \in L$. The links and constraints are defined as follows:

$$l^i_{p,p+1} = (n^i_j, n^i_{j+1}) \cup (\tau(i), \Lambda^O(i)) \cup (\Lambda^O(i), \Lambda^F(i)) \quad for\ j \in [k-1] \tag{3}$$

$$t_i^p + T^{min}_{il^i_{p,p+1}} \leq t_i^{p+1} \leq t_i^p + T^{max}_{il^i_{p,p+1}} \tag{4}$$

C3: Definition of immediate predecessors x^n_{ij}:

$$\sum_{i\in A^n_0} x^n_{ij} = 1 \quad \forall n \in N, \forall j \in A^n_0 \tag{5a}$$

$$\sum_{j\in A^n_0} x^n_{ij} = 1 \quad \forall n \in N, \forall i \in A^n_0 \tag{5b}$$

$$t^n_j \geq t^n_i - (1 - x^n_{ij})M \quad \forall n \in N, \forall i, j \in A^n, i \neq j \tag{5c}$$

$$t^n_i \geq t^n_j - (1 - x^n_{0j})M \quad \forall n \in N, \forall i, j \in A^n, i \neq j \tag{5d}$$

$$t^n_i \geq t^n_j - (1 - x^n_{i0})M \quad \forall n \in N, \forall i, j \in A^n, i \neq j \tag{5e}$$

C4: Definition of predecessors y^n_{ij}:

$$y^n_{ij} \geq x^n_{ij} \quad \forall n \in N, \forall i, j \in A^n_0, i \neq j \tag{6a}$$

$$y^n_{ij} + y^n_{ji} = 1 \quad \forall n \in N, \forall i, j \in A^n, i \neq j \tag{6b}$$

$$y^n_{ji} \geq x^n_{0j} \quad \forall n \in N, \forall i, j \in A^n, i \neq j \tag{6c}$$

$$x^n_{0j} + y^n_{ij} \leq 1 \quad \forall n \in N, \forall i, j \in A^n, i \neq j \tag{6d}$$

$$y^n_{ji} \geq x^n_{i0} \quad \forall n \in N, \forall i, j \in A^n, i \neq j \tag{6e}$$

$$x^n_{i0} + y^n_{ij} \leq 1 \quad \forall n \in N, \forall i, j \in A^n, i \neq j \tag{6f}$$

$$y^n_{kj} \geq x^n_{ij} + y^n_{ki} - 1 \quad \forall n \in N, \forall i, j, k \in A^n, i \neq j \neq k \tag{6g}$$

C5: The following constraint prevents overtaking:

$$y^n_{ij} - y^m_{ij} = 0 \quad \forall i, j \in A, i \neq j \quad , \forall l_{n,m} \in (P_i \cup \Lambda(i)) \cap (P_j \cup \Lambda(j)) \tag{7}$$

C6: For preventing head-on collision of two VTOL on a link $l_{n,m}$, the following constraint is used:

$$y^n_{ij} - y^m_{ij} = 0 \quad \forall i, j \in A, i \neq j \quad \forall l_{n,m} \in (P_i \cup \Lambda(i))\, and\ l_{m,n} \in (P_j \cup \Lambda(j)) \tag{8}$$

C7: This constraint maintains the separation requirement on the taxi and surface direction, where L^{sep}_{ij} is the separation requirement between VTOL in units of distance over the edge (or link) length of $L(n, m)$.

$$t^n_j \geq t^n_i + \frac{L^{sep}_{ij}}{L(n,m)}(t^m_i - t^n_i) - (1 - y^n_{ij})M \quad \forall l_{n,m} \in L_T \cup L_F \ \forall i, j \in A^n, i \neq j \tag{9}$$

C8: Similarly, constraint equation (10) maintains the wake vortex requirement on the TLOF pad when the VTOL i completes a takeoff where W^{tsep}_{ij} is the time VTOL j should wait after the departure of VTOL i.

$$t^r_j \geq t^r_i + W^{tsep}_{ij} - (1 - x^r_{ij})M \quad \forall i, j \in A^n, i \neq j, \tau(i) = \tau(j) = r \tag{10}$$

C9: Constraint equations (11) and(4) maintain the time continuity over the VTOL's route.

$$t_i^{\tau(i)} \geq t_i^{n^i_{k_i}} \quad \forall i \in A \tag{11}$$

C10: After the departure of a VTOL from the assigned TLOF pad, its immediate follower cannot enter till the departing VTOL has crossed its OFV boundary. Constraint (12) ensure this.

$$t^r_j \geq t_i^{\Lambda^O(i)} - (1 - x^r_{ij})M \quad \forall i, j \in A, i \neq j, \tau(i) = r = \tau(j) \tag{12}$$

C11: Binary and non-negativity constraints:

$$t^n_i \in R^+ \forall i \in A, \forall n \in N \tag{13a}$$

$$x^n_{ij}, y^n_{ij} \in \{0, 1\} \forall n \in N, \forall i, j \in A^n_0, i \neq j \tag{13b}$$

Fairness: C12: We impose constraints to ensure fairness of the delay incurred by each VTOL. We restrict the delay of each VTOL within a specified percentage P of the mean delay of all VTOL. We constrain either Total Delay (14a) or Gate Delay (14b), i.e. time spent at the gate before taxiing.

$$t_i^{\Lambda^F(i)} - \mathtt{DRGATE_i} \leq (1+P)\frac{\sum_{k\in A}(t_k^{\Lambda^F(k)} - \mathtt{DRGATE_k})}{|A|}, \forall i \in A \tag{14a}$$

$$t_i^{n_1^i} - \mathtt{DRGATE_i} \leq (1+P)\frac{\sum_{k\in A}(t_k^{n_1^k} - \mathtt{DRGATE_k})}{|A|}, \ \forall i \in A \tag{14b}$$

IV. Experiment Setup

We used MATLAB version 2022a (9.12) with an optimization toolbox of version 9.3 on a Dell Optiplex 5060 Desktop PC equipped with an i5-8500 processor having 6 cores @ 3GHz and 16GB RAM, to solve our optimization problem. Our implementation is available on GitHub, which can be used to reproduce results [20].

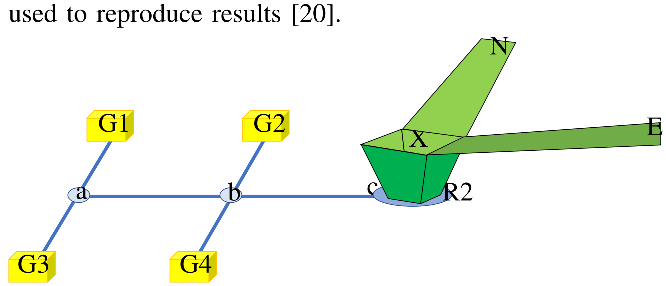

Fig. 4: Sample topology with 4 gates and 2 surface directions.

Figure 4 shows the topology in our experimental setup with 2 surface directions. All taxi edges in the network are

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equal in length and similarly, all surface direction edges are equal. However, the length of the surface direction edges is greater than that of the taxi edges. We consider five categories of VTOLs, with the assumption that the taxiing speed and climbing speed are the same for all UAM classes. However, distance separation and wake vortex time requirements vary among VTOL classes. In our configuration, TLOF pad R2 is restricted for departures and therefore eliminates head-on collisions at every link. We randomly chose the VTOL class, starting gate and $\mathtt{DRGATE_i}$. The weights in Table II are set as $W_g = 0.2, W_t = 0.8, W_c = 0.7$. The large constant M is set as 600. We utilise the optimisation problem described in Section III-A to obtain the time by which a VTOL reaches a node. We calculate the time it takes on its route when no other VTOL is present, called $ZeroDelayTime$. By comparing the actual time with its $ZeroDelayTime$, we can determine the delay over its departure route.

Experiment 1: Common line segment. We will demonstrate the impact of reduction in length of the shared path on traffic congestion. We define a variable $\rho = \frac{Length('R2-X-N')}{Length('R2-X')}$. We will use only 1 surface direction, 'R2-X-N', altering its length according to the length of 'R2-X' by changing ρ value from 15 to 1 in variable step size. We record the delay incurred by all flights for each value of ρ and then compare the results to determine the optimal value of ρ that results in minimum delay for the departing flights. The total number of runs were 24 (6 ρ and 4 different no. of flights).

Experiment 2: Multiple surface directions. Our aim is to identify the optimal number of surface directions that might result in minimum delay. We considered a total of 4 surface directions from 'X'. Each run of the experiment comprised of varying number of flights between 5 and 20, with $\rho = 5$. The total number of runs was 16 (4 different no. of flights and 4 directions).

Experiment 3: Fairness. To analyse fairness metrics with total delay (equation (14a)) and gate delay (equation (14b)). In this experiment, we fix the number of flights to 15, 4 surface directions and $\rho = 5$. The total number of runs was 6 (3 P values for 'Gate Delay' and 'Total Delay' constraints). Table IV shows the $\mathtt{DRGATE_i}$ and the starting gate of each flight. In each column, first row is the index of a flight, the second row is the $\mathtt{DRGATE_i}$ time in seconds and the third row is the starting gate of the flight. As an example, flight index 9 will initiate taxiing 12 seconds from 'G3' gate.

V. Results

Figure 5 shows the increase in delay when the length of the common path increases (experiment 1). The results are plotted for a varying number of flights. As expected, reducing the common path length from ρ=15 (longest) to ρ=1 (shortest) leads to a decrease in delay for each flight. The reason is that flights can turn in different directions after reaching the end of the path, and separation constraints are not required to be maintained. On average, reducing the common path length ratio from 15 to 1 results in over 50% reduction in delay.

Figure 6 shows the reduction of common path length by the introduction of multiple surface directions at the OFV boundary (experiment 2). For a given surface direction, the average delay increases with an increase in the number of flights. However, as expected, increasing the number of surface directions decreases the average delay. For the two successive VTOLs taking different climb directions after the take-off, the immediate follower can take-off by adhering only to wake vortex separation, reducing its overall delay. Like the runway, the TLOF pad remains a bottleneck for VTOL vehicles, however, only up to the OFV boundary.

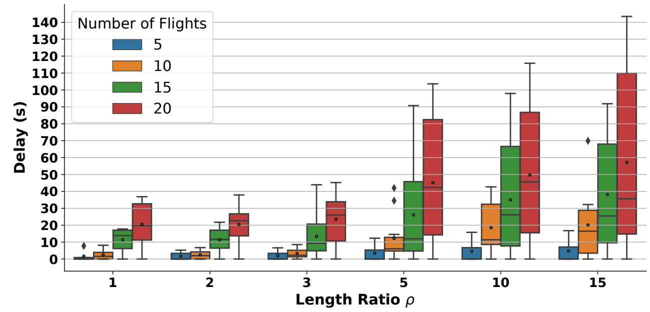

Fig. 5: Results of common line segment (experiment 1) for 5,10,15,20 flights indicated by different colours. The mean value of each box plot is marked in a small circle.

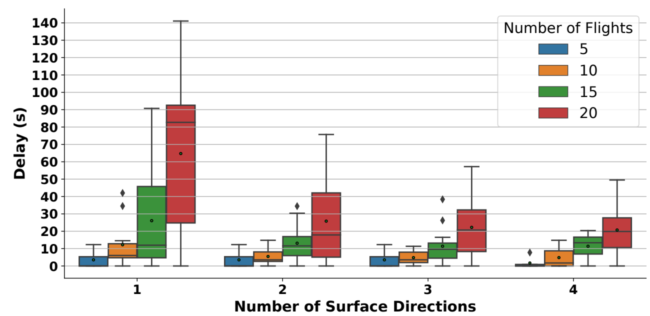

Fig. 6: Results of Multiple surface direction (experiment 2).

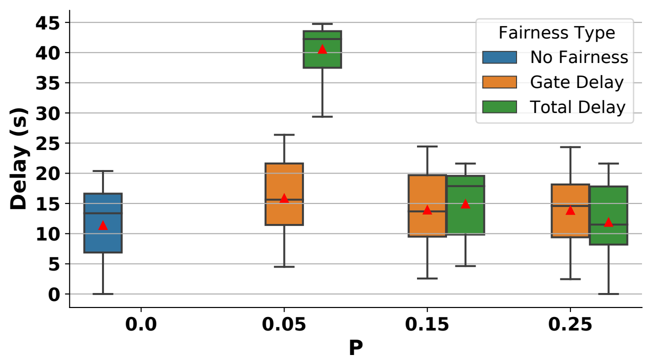

Fig. 7: Results of Fairness (experiment 3).

As discussed in Section III, there are 3 significant delays in the path of a VTOL: (a) delay on the taxiway; (b) delay on TLOF pad and OFV; (c) delay on the climb surface direction. Waiting at Gate before entering a taxiway introduces the 4^{th} delay, called gate delay. Without any fairness constraint, the total delay is equal to the gate delay. This is because waiting at the gate has a minimum weight (W_g) than the other weights. This causes all the flights to remain at the gate rather than slowing down on the taxiway or surface direction, preventing congestion. Figure 7 shows a delay comparison with the presence and absence of fairness constraint (experiment 3). The delay range has shifted to a higher side with

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Flight Index (i)	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
$\mathtt{DRGATE_i}$ (s)	19	32	47	48	53	12	17	30	45	12	11	33	17	46	17
Gate (n_1^i)	G4	G3	G4	G2	G2	G4	G3	G1	G2	G3	G4	G4	G1	G4	G1

TABLE IV: The $\mathtt{DRGATE_i}$ of each VTOL for experiment 3

the fairness constraint. For a small value of P, the shift in 'Total Delay' is significant than the 'Gate Delay'. This shift is because the optimisation solver adds a substantial delay to each VTOL to limit deviations from the mean. However, for a higher value of P, the shift becomes insignificant, and the case is comparable to the one without fairness. Thus, the optimal value of P lies between 10% to 20%. Figure 8 shows the distribution of the 4 delays discussed earlier within the total delay for the fairness scenario presented in Figure 7 for three P values. The case labelled as "Total | P = 0.25" can be considered as a base case comparable to the "No fairness" case, where total delay is equal to gate delay. We can observe that restricting the 'Gate delay' constraint (see equation (14b)) increases the share of taxi delay while restricting the 'Total delay' constraint (see equation (14a)) slightly increases the share of climb delay. In case of restricting the Gate delay, all VTOLs will wait for, on an overage, the same amount of time and then enter a taxiway. This causes congestion on the taxiways, and hence taxi delay increases. This can be clearly observed in the case labelled "Gate | P= 0.05". When we restrict Total Delay, the optimisation solution tends to allocate most of the delay to gate delay and the remainder to climb delay, consequently reducing ground congestion. In practice, a VTOL can reduce its climb delay if the regulation permits, as it is the last part of the path of a VTOL, but the same can't be said for taxiing, as it may affect the free flow of traffic on the ground.

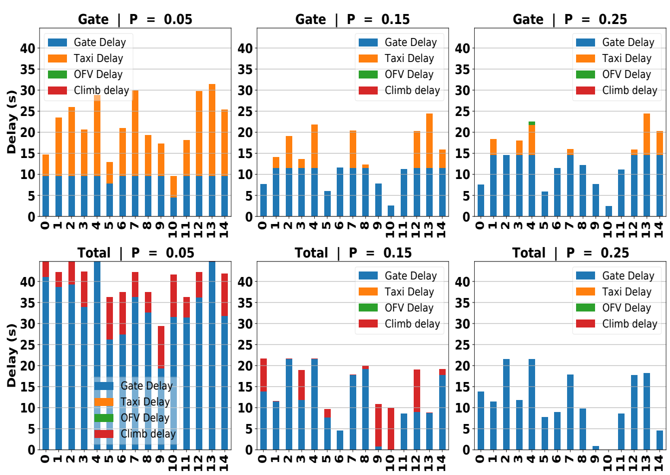

Fig. 8: Delay distribution of each flight when fairness policy uses:(a) 'Gate delay' constraint and (b) 'Total delay' constraint for different values of P.

VI. Conclusion

This paper proposes and implements a novel approach that introduces multiple climb directions for VTOL vehicles and demonstrates the efficacy of this approach to reduce congestion on vertiminal. The results clearly show that increasing the number of surface directions reduces the overall delay of flights by more than 50%, as adding more directions effectively reduces the shared path length. To ensure fairness among the VTOLs, there should be restrictions on the Total delay incurred by each flight to be not more than 10% to 20% of the mean delay value. On a vertiminal having multiple TLOF pads, each TLOF pad can have multiple surface directions, thereby enabling a significant increase in traffic throughput. However, it is essential to consider economic costs when deciding whether to implement this approach. Overall, this work opens up new possibilities for optimising the operation of VTOL vehicles in UAM systems.

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